

# CHECKPOINT TRAJECTORY FORECASTING: WHEN EARLY TRAINING DYNAMICS FALL SHORT OF PREDICTING FINAL MODEL QUALITY

**Anonymous authors**

Paper under double-blind review

## ABSTRACT

Neural network training generates sequences of intermediate weight checkpoints, yet most weight space learning methods analyze only final trained models. We investigate Checkpoint Trajectory Forecasting (CTF), a framework that treats sequences of training checkpoints as temporal data to predict final model properties from early training stages. Our approach embeds each checkpoint using statistical weight features, then processes the sequence with temporal models (LSTM or Transformer) to forecast final test accuracy. Experiments across MNIST, Fashion-MNIST, and CIFAR-10 reveal that while CTF achieves high Spearman correlation ( $\rho = 0.93$ ) on combined datasets, this success is largely attributable to dataset-level separability rather than fine-grained within-dataset ranking. Critical ablations show that weight normalization and rich feature extraction are essential—removing normalization causes complete failure ( $\rho = 0.00$ ), and reducing features from 8 to 2 per layer drops correlation from 0.84 to 0.13. Surprisingly, temporal ordering provides minimal benefit, with shuffled checkpoints achieving comparable performance. These findings highlight significant limitations in trajectory-based forecasting for practical hyperparameter selection.

## 1 INTRODUCTION

Training neural networks is computationally expensive, and practitioners frequently face decisions about which training runs to continue and which to terminate early. The ability to predict final model properties from early training stages could enable significant computational savings in hyperparameter search and neural architecture search. This motivates Checkpoint Trajectory Forecasting (CTF): using sequences of intermediate weight checkpoints to forecast properties of the final trained model.

Existing weight space learning methods primarily analyze single trained models as static snapshots (Schürholt et al., 2024; Kahana et al., 2024), ignoring the rich temporal structure present during training. Model zoo datasets contain only final trained models (Schürholt et al., 2022; Falk et al., 2025a), and while some recent work models weight trajectories (Shou et al., 2025; Gupta et al., 2026), these focus on predicting final weights rather than model properties. Learning curve prediction methods (Lin et al., 2024; Solís-Martín et al., 2025) use training loss curves but do not leverage weight-space information.

We hypothesize that the sequence of weight checkpoints during early training contains predictive signals about final model quality that can be extracted using sequence models. To test this, we construct checkpoint trajectory datasets by training MLPs on MNIST, Fashion-MNIST, and CIFAR-10 with diverse hyperparameters, saving checkpoints at each epoch. We embed checkpoints using statistical weight features and train LSTM (Hochreiter & Schmidhuber, 1997) and Transformer (Vaswani et al., 2017) models to predict final test accuracy from the first 20% of training.

## 2 RELATED WORK

**Weight Space Learning.** Recent work has developed methods for learning from neural network weights directly. SANE (Schürholt et al., 2024) processes weight subsets as tokens to learn task-agnostic representations, while ProbeGen (Kahana et al., 2024) represents models by passing learned

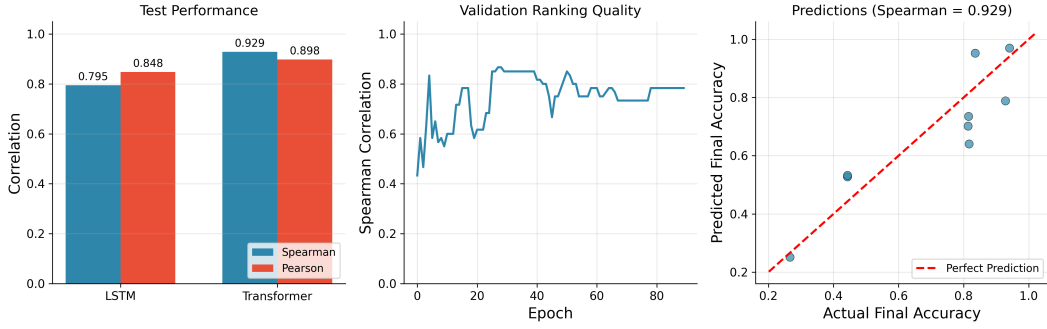


Figure 1: Main results on combined dataset. (Left) Test performance comparison: Transformer achieves Spearman  $\rho = 0.929$  vs. LSTM’s  $\rho = 0.795$ , with Transformer also achieving higher Pearson correlation (0.898 vs. 0.848). (Center) Validation ranking quality during Transformer training shows substantial variance. (Right) Scatter plot of predictions vs. actual accuracy; deviations in the mid-range suggest difficulty with within-dataset ranking.

probes through them. These methods analyze single trained models rather than training trajectories. Weight space augmentation techniques (Shamsian et al., 2024) address overfitting challenges but do not consider temporal dynamics. Model zoo datasets (Schürholt et al., 2022; Falk et al., 2025a;b) provide diverse collections of trained neural networks but contain only final models.

**Training Dynamics and Learning Curves.** Loss landscape visualization (Li et al., 2017) examines the geometry of neural network optimization but does not leverage sequential checkpoint information. Gradient Flow Matching (Shou et al., 2025) and DeepWeightFlow (Gupta et al., 2026) model weight trajectories but focus on generating final weights rather than predicting model properties. Methods for predicting training outcomes from partial curves (Lin et al., 2024; Solís-Martín et al., 2025) typically use loss or accuracy values over time rather than weight-space features.

### 3 METHOD

**Problem Setup.** Given a sequence of weight checkpoints  $\{W_1, W_2, \dots, W_k\}$  from the first  $k$  epochs of training (where  $k$  represents 20% of total training), we aim to predict the final test accuracy  $y$  of the trained model.

**Weight Embedding.** Each checkpoint  $W_t$  is embedded into a fixed-size vector  $z_t \in \mathbb{R}^d$  using statistical features. For each weight matrix in the network, we compute 8 statistics: mean, standard deviation, L2 norm, minimum, maximum, 25th percentile, 75th percentile, and mean absolute value. These are concatenated across layers to form the checkpoint embedding. We normalize embeddings across the dataset to have zero mean and unit variance.

**Temporal Model.** The sequence of embeddings  $\{z_1, \dots, z_k\}$  is processed by either an LSTM or Transformer encoder. For the LSTM, we use a 2-layer architecture with hidden dimension 64 and dropout 0.3, taking the final hidden state as the sequence representation. For the Transformer, we project embeddings to dimension 64, add learned positional encodings, and apply 2 encoder layers with 4 attention heads. The final token representation is passed through an MLP to predict the target property.

### 4 EXPERIMENTS

**Setup.** We generate 60 training trajectories (20 per dataset) by training 3-layer MLPs on MNIST, Fashion-MNIST, and CIFAR-10. Hyperparameters are sampled from: hidden sizes  $\{32, 64, 128, 256\}$ , learning rates  $\{0.0001, 0.0005, 0.001, 0.003, 0.01\}$ , and weight decay  $\{0, 0.001\}$ . Each model is trained for 25 epochs with checkpoints saved every epoch. Trajectories are split 70%/15%/15% for train/validation/test.

Table 1: Dataset statistics and per-dataset CTF performance. The large between-dataset variance in accuracy (combined std = 0.254) dominates within-dataset variance (0.011–0.085), enabling easy dataset separation and inflating combined correlation.

| Dataset       | Mean Acc. | Std Acc. | Spearman $\rho$ |
|---------------|-----------|----------|-----------------|
| MNIST         | 0.936     | 0.011    | 0.21            |
| Fashion-MNIST | 0.816     | 0.014    | 0.40            |
| CIFAR-10      | 0.354     | 0.085    | 0.86            |
| Combined      | 0.702     | 0.254    | 0.93            |

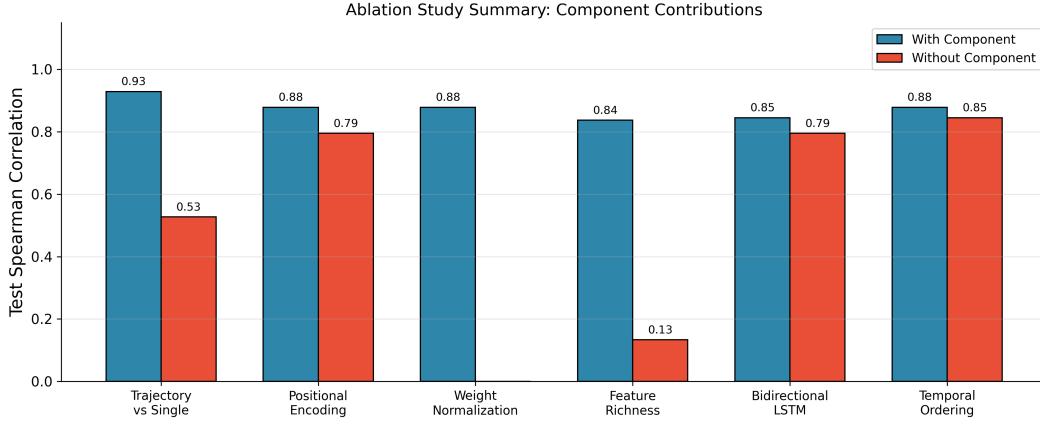


Figure 2: Ablation study summary showing Test Spearman correlation with and without each component. Weight normalization is critical (0.88  $\rightarrow$  0.00), feature richness matters substantially (0.84  $\rightarrow$  0.13), temporal ordering provides minimal benefit (0.88  $\rightarrow$  0.85), and full trajectories outperform single checkpoints (0.93  $\rightarrow$  0.53).

**Main Results.** Figure 1 shows that the Transformer model achieves Spearman correlation of 0.929 compared to 0.795 for LSTM on the combined test set. The Transformer also achieves higher Pearson correlation (0.898 vs. 0.848). The validation curve shows high variance even in later epochs, indicating training instability. However, this strong overall performance must be interpreted carefully given the dataset composition.

**The Dataset Separability Problem.** Table 1 reveals a critical limitation. The three datasets have dramatically different accuracy ranges, and the combined dataset standard deviation (0.254) is dominated by between-dataset variance rather than within-dataset variance (0.011–0.085). Per-dataset correlations are substantially lower: MNIST achieves only  $\rho = 0.21$  and Fashion-MNIST  $\rho = 0.40$ , suggesting the model excels at dataset identification but struggles with fine-grained ranking where it matters most for hyperparameter selection. Only CIFAR-10, with its broader accuracy distribution, achieves reasonable within-dataset correlation ( $\rho = 0.86$ ).

**Ablation Analysis.** Our ablations (Figure 2) reveal several critical findings. Weight normalization is essential: without it, Spearman correlation drops from 0.88 to 0.00, as raw weight magnitudes vary dramatically across configurations. Reducing features from 8 per layer to 2 (mean and standard deviation only) drops correlation from 0.84 to 0.13, demonstrating that higher-order statistics capture important distributional information. Surprisingly, shuffling the temporal order of checkpoints has minimal effect ( $\rho = 0.88$  ordered vs.  $\rho = 0.85$  shuffled), suggesting the model relies more on aggregate statistics than temporal dynamics. Using the full trajectory ( $\rho = 0.93$ ) substantially outperforms using only the final early checkpoint ( $\rho = 0.53$ ), though this may reflect ensemble-like averaging rather than temporal pattern recognition.

## 5 CONCLUSION

We investigated Checkpoint Trajectory Forecasting for predicting final model properties from early training checkpoints. While achieving high overall correlation ( $\rho = 0.93$ ), our analysis reveals this success is largely attributable to dataset-level separability rather than fine-grained ranking ability. Critical findings include: (1) weight normalization is absolutely essential, (2) rich statistical features outperform simple moments, and (3) temporal ordering provides surprisingly little benefit. These results suggest that while checkpoint trajectories contain some predictive signal, the practical utility for hyperparameter selection within a fixed task may be limited. Future work should focus on within-dataset evaluation and investigate whether different target properties are more amenable to trajectory-based prediction.

## REFERENCES

- Damian Falk, Léo Meynert, Florence Pfammatter, Konstantin Schürholt, and Damian Borth. A model zoo of vision transformers. *ArXiv*, abs/2504.10231, 2025a.
- Damian Falk, Konstantin Schürholt, and Damian Borth. The impact of model zoo size and composition on weight space learning. *ArXiv*, abs/2504.10141, 2025b.
- Saumya Gupta, Scott Biggs, Moritz Laber, Zohair Shafi, R. Walters, and Ayan Paul. Deepweight-flow: Re-basined flow matching for generating neural network weights. *ArXiv*, abs/2601.05052, 2026.
- Sepp Hochreiter and J. Schmidhuber. Long short-term memory. *Neural Computation*, 9:1735–1780, 1997.
- Jonathan Kahana, Eliahu Horwitz, Imri Shuval, and Yedid Hoshen. Deep linear probe generators for weight space learning. *ArXiv*, abs/2410.10811, 2024.
- Hao Li, Zheng Xu, Gavin Taylor, and T. Goldstein. Visualizing the loss landscape of neural nets. *ArXiv*, abs/1712.09913, 2017.
- J. Lin, S. Ament, M. Balandat, and E. Bakshy. Scaling gaussian processes for learning curve prediction via latent kronecker structure. *ArXiv*, abs/2410.09239, 2024.
- Konstantin Schürholt, Diyar Taskiran, Boris Knyazev, Xavier Gir’o i Nieto, and Damian Borth. Model zoos: A dataset of diverse populations of neural network models. *ArXiv*, abs/2209.14764, 2022.
- Konstantin Schürholt, Michael W. Mahoney, and Damian Borth. Towards scalable and versatile weight space learning. *ArXiv*, abs/2406.09997, 2024.
- Aviv Shamsian, Aviv Navon, David W. Zhang, Yan Zhang, Ethan Fetaya, Gal Chechik, and Haggai Maron. Improved generalization of weight space networks via augmentations. pp. 44378–44393, 2024.
- Xiao Shou, Yanna Ding, and Jianxi Gao. Gradient flow matching for learning update dynamics in neural network training. *ArXiv*, abs/2505.20221, 2025.
- David Solís-Martín, Juan Galán-Páez, and J. Borrego-Díaz. A model for learning-curve estimation in efficient neural architecture search and its application in predictive health maintenance. *Mathematics*, 2025.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and I. Polosukhin. Attention is all you need. pp. 5998–6008, 2017.

## SUPPLEMENTARY MATERIAL

### A EXPERIMENTAL DETAILS

**Model Architecture.** The base MLP used for generating trajectories has architecture:  $\text{Linear}(\text{input\_dim}, \text{hidden}) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.2) \rightarrow \text{Linear}(\text{hidden}, \text{hidden}/2) \rightarrow \text{ReLU} \rightarrow \text{Linear}(\text{hidden}/2, 10)$ . Input dimensions are 784 for MNIST/Fashion-MNIST and 3072 for CIFAR-10.

**CTF Model Details.** The LSTM uses 2 layers with hidden dimension 64 and dropout 0.3. The Transformer uses 2 encoder layers with 4 attention heads, hidden dimension 64, and feedforward dimension 128. Both models use a 2-layer MLP head:  $\text{Linear}(64, 32) \rightarrow \text{ReLU} \rightarrow \text{Dropout} \rightarrow \text{Linear}(32, 1)$ .

**Training Protocol.** CTF models are trained for up to 150 epochs with early stopping (patience 30) based on validation loss. We use AdamW with learning rate 0.001 and weight decay 0.01. Learning rate is reduced by factor 0.5 when validation loss plateaus for 10 epochs. Gradient norms are clipped to 1.0.

### B DATASET ANALYSIS

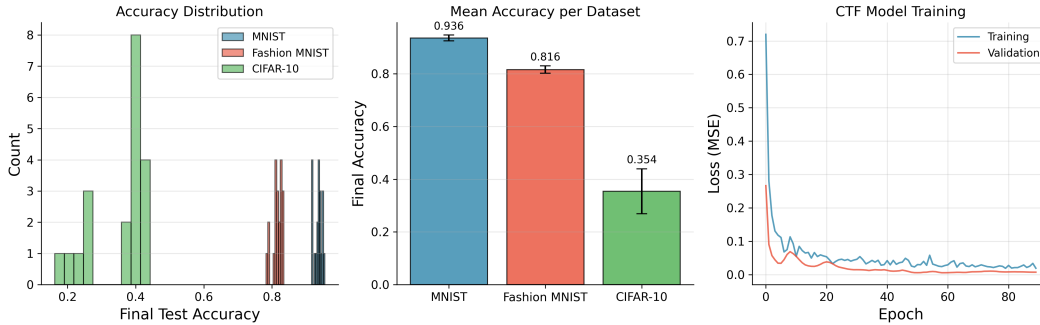


Figure 3: Dataset characteristics. (Left) Accuracy distributions show MNIST and Fashion-MNIST cluster at high accuracies while CIFAR-10 shows a broader distribution centered around 0.35. (Center) Mean accuracies with standard deviations confirm large between-dataset variance. (Right) CTF model training loss converges smoothly.

The accuracy distributions in Figure 3 illustrate the dataset separability problem discussed in the main text. MNIST and Fashion-MNIST cluster at high accuracies with very low variance, making within-dataset ranking extremely challenging.

### C EXTENDED ABLATION RESULTS

### D ARCHITECTURE ABLATIONS

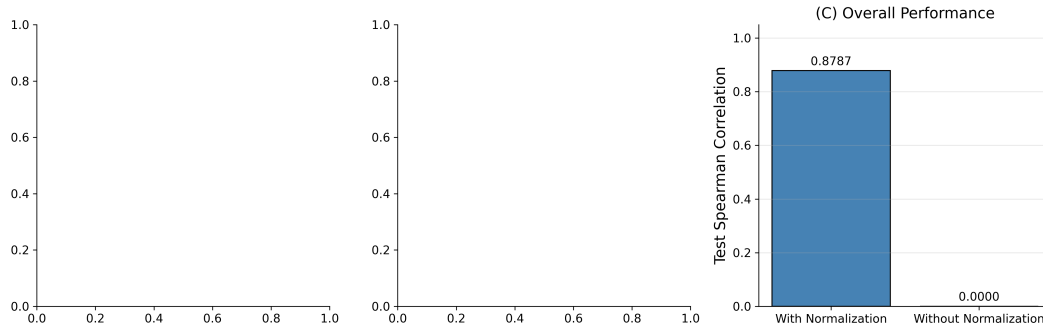


Figure 4: Weight normalization ablation. (A) Test correlations with vs. without normalization. (B) Training dynamics reveal failure without normalization. (C) Scatter plot confirms complete prediction failure without normalization.

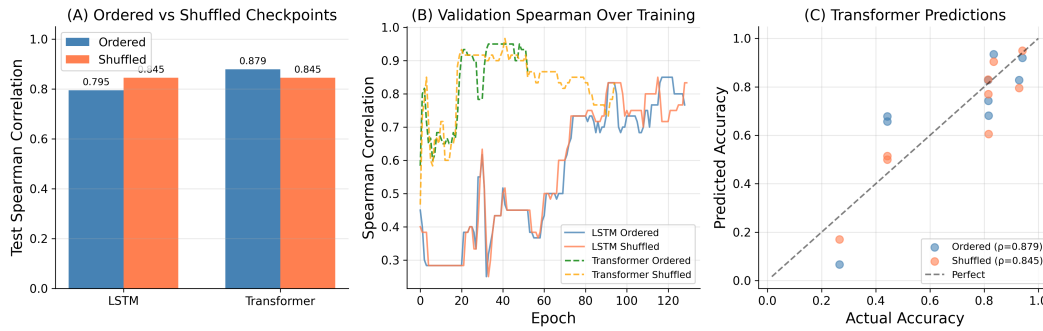


Figure 5: Temporal ordering ablation. (A) Ordered vs. shuffled checkpoint sequences. (B) Validation correlation during training. (C) Scatter plot of predictions. Neither model strongly relies on temporal structure.

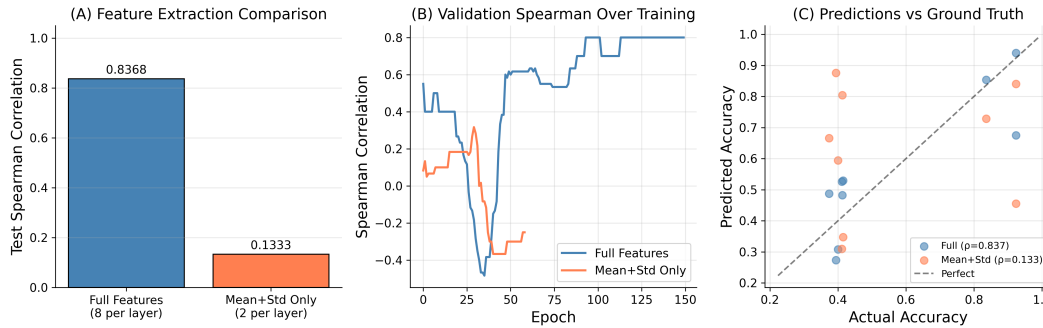


Figure 6: Statistical features ablation. (A) Full 8-feature statistics vs. mean and std only. (B) Training dynamics show reduced features fail to learn. (C) Scatter plot confirms poor prediction with reduced features (0.837  $\rightarrow$  0.133).

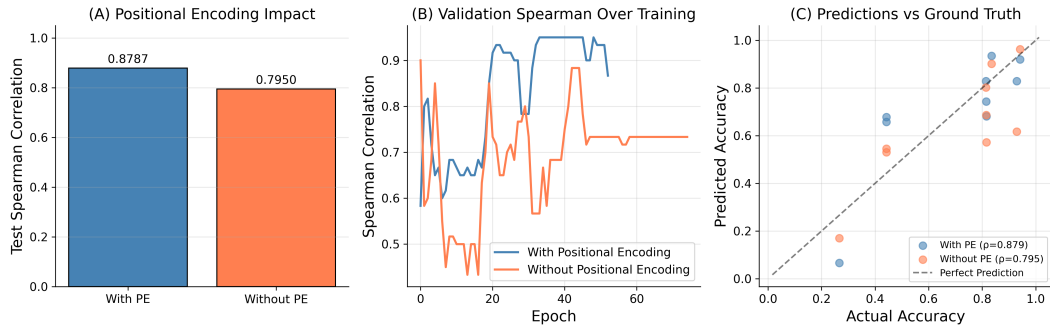


Figure 7: Positional encoding ablation for Transformer. Adding positional encoding improves Spearman correlation from 0.795 to 0.879.

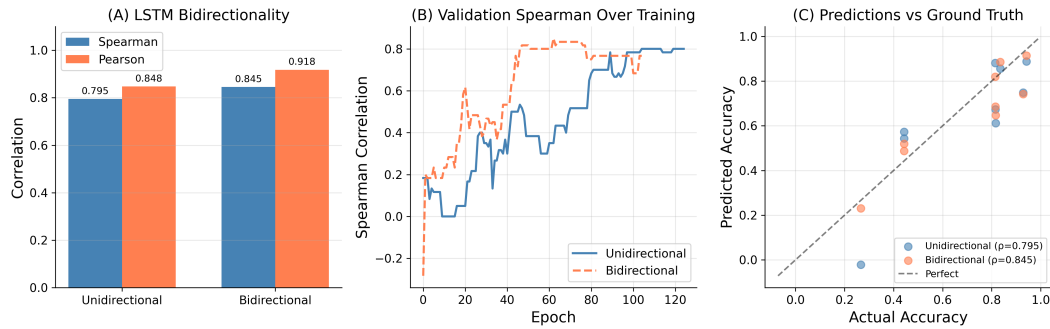


Figure 8: Bidirectional LSTM ablation. Bidirectional processing improves Spearman from 0.795 to 0.845 and Pearson from 0.848 to 0.918.

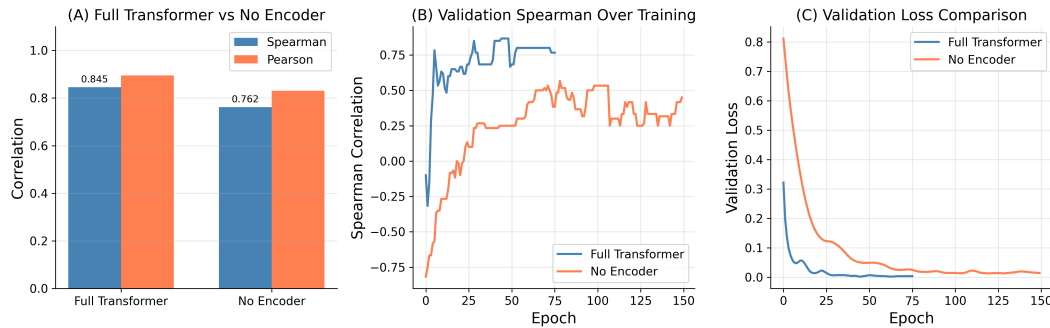


Figure 9: Ablation removing the temporal encoder entirely, using only averaged checkpoint embeddings. The full Transformer outperforms the no-encoder baseline, confirming the sequence model provides value beyond simple averaging.